



Karnataka State Police
Government of Karnataka

Powered by **H2S**

0atathon 2026

Technology partner



Team Details :-

- **Team name : JayaDhwaja**
- **Team leader name: Jashwanth Ramadugu**
- **Team size : 2**
- **Challenge : AI-Driven Crime Analytics and Visualization Platform**



BRIEF ABOUT THE SOLUTION :-

- ❖ Our proposed solution, the **Crime Intelligence & Analytical Platform**, transforms raw crime records into meaningful insights using data analytics, interactive visualizations, and machine learning.

The platform enables law enforcement agencies to analyze historical crime patterns, identify crime hotspots, monitor district-wise trends, and forecast future crime risks. By replacing manual spreadsheet-based analysis with an intelligent dashboard, it supports proactive policing, efficient resource allocation, and faster decision-making.

The solution integrates **Excel, SQL, Power BI, Machine Learning, and Flask** to provide a user-friendly web interface where officers can explore crime statistics, visualize trends, and access predictive analytics in real time.

KEY OBJECTIVES :-

- Identify Crime Hotspots
- Analyze Historical Crime Trends
- Predict Future Crime Risk
- Support Data-Driven Policing
- Improve Decision-Making



Opportunities :-

- **How different is it from existing solutions?**

Integrates **Analytics, Visualization, and AI Prediction** into a single platform.
Converts static crime records into **interactive insights** for faster analysis.
Designed specifically to support **law enforcement decision-making**.

- **How will it solve the problem?**

Identifies crime hotspots and emerging trends.
Predicts future crime patterns using Machine Learning.
Enables proactive policing through real-time dashboards and insights.

- **USP of the Proposed Solution:-**

AI-powered crime prediction for preventive policing.
Interactive Power BI dashboards for quick analysis.
Web-based platform with an intuitive interface.
Data-driven resource allocation to improve operational efficiency.



List of Features Offered by the Solution :-

➤ **Interactive Crime Dashboard**

Visualizes district-wise crime statistics, trends, and key performance indicators.

➤ **Crime Hotspot Analysis**

Identifies high-risk locations using historical crime data.

➤ **Trend Analysis**

Analyzes crime patterns across different years and crime categories.

➤ **AI-Based Crime Prediction**

Forecasts future crime trends using Machine Learning models.

➤ **Advanced Data Filtering**

Filter records by district, year, crime category, and other parameters.

➤ **User-Friendly Web Platform**

Provides a simple Flask-based interface for accessing dashboards and insights.

➤ **Automated Reporting**

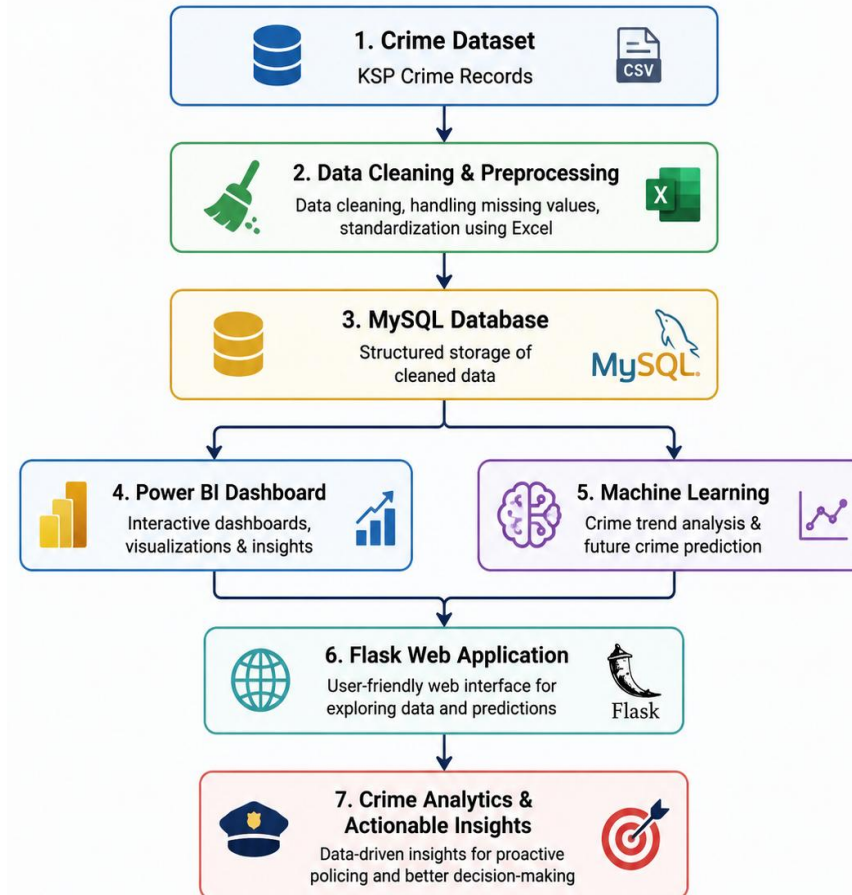
Generates structured insights to support faster decision-making.

➤ **Decision Support System**

Helps law enforcement plan preventive measures and optimize resource allocation.



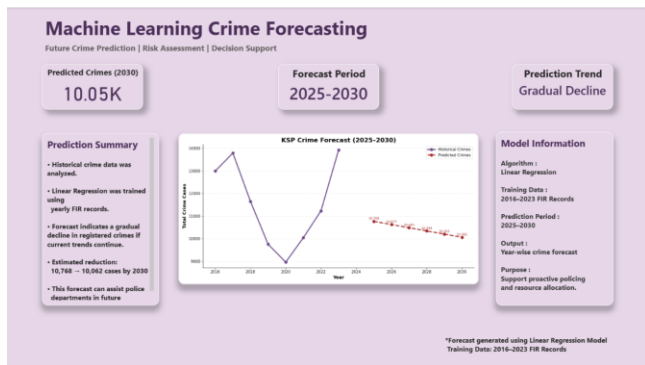
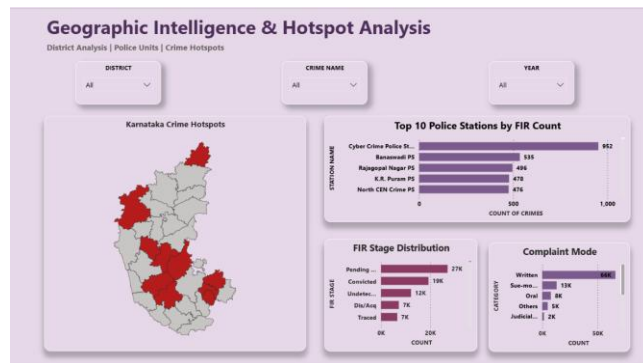
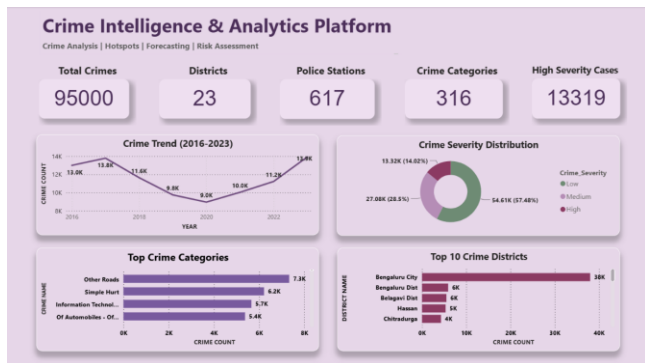
Process flow diagram :-





Wireframes/Mock diagrams of the proposed solution:-

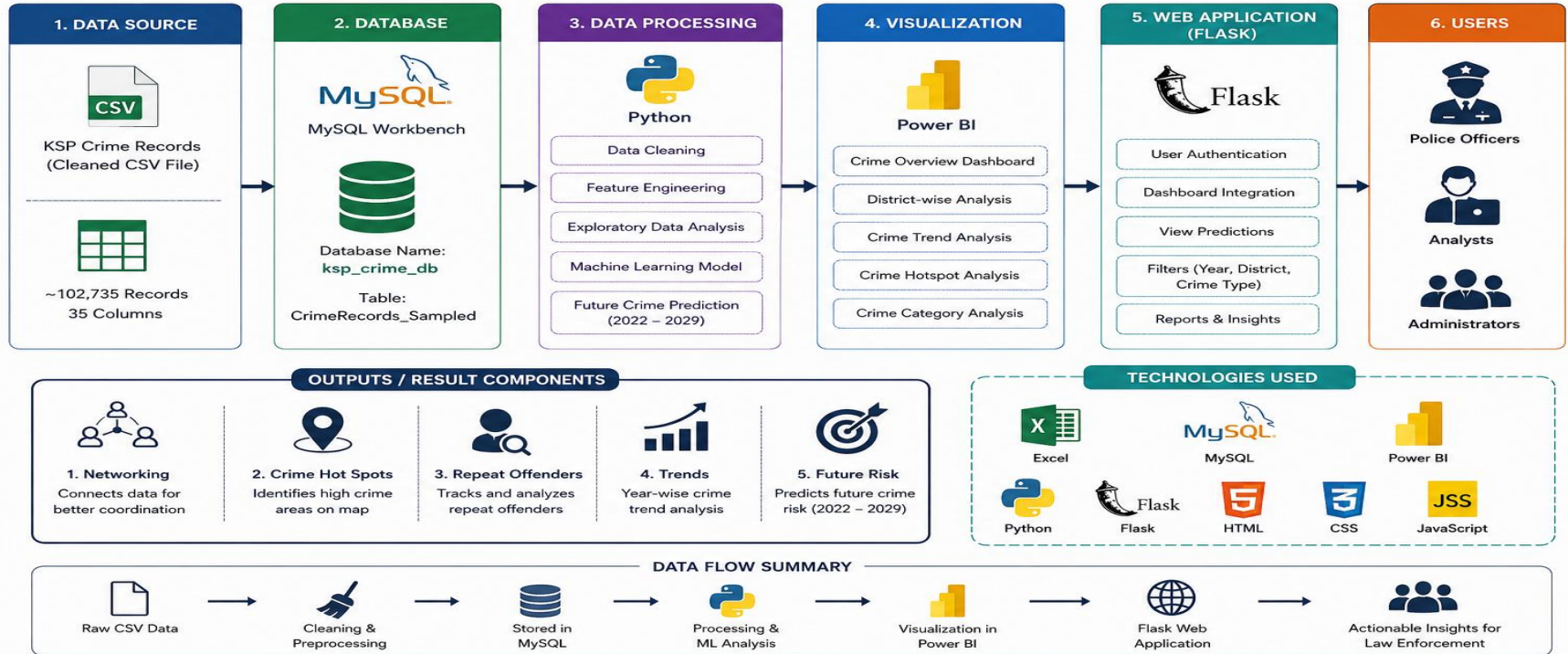
- The following wireframes illustrate the user interface and navigation flow of the proposed Crime Intelligence & Analytical Platform.



- The Crime Intelligence & Analytical Platform leverages Data Analytics, Power BI, SQL, Machine Learning, and Flask to transform crime records into actionable insights, enabling smarter and proactive policing.



Architecture diagram of the proposed solution :-





Technologies used in the solution :-

Category	Technology	Purpose
Data Preparation	Microsoft Excel	Data Cleaning & Preprocessing
Database	MySQL Workbench	Structured Crime Data Storage
Query Language	SQL	Data Retrieval & Analysis
Programming	Python	Data Processing & Machine Learning
Data Visualization	Power BI	Interactive Dashboards & Reports
Machine Learning	Scikit-learn	Crime Trend Prediction
Web Framework	Flask	Web Application Development
Frontend	HTML, CSS, JavaScript	User Interface Development



List down Catalyst Services being used in the solution :-

Catalyst Service	Purpose
AppSail	Hosting & Deployment of the Flask Web Application
Catalyst CLI	Application Initialization & Deployment Management
Development Environment	Application Testing & Deployment Workflow



Estimated implementation cost :-

Component	Technology Used	Estimated Cost
Data Preparation	Microsoft Excel	Free (Student License)
Database	MySQL Community Edition	Free
Programming	Python	Free (Open Source)
Machine Learning	Scikit-learn	Free (Open Source)
Dashboard	Power BI Desktop	Free
Web Development	Flask	Free (Open Source)
Deployment	Zoho Catalyst AppSail	Free Tier
Development Tools	VS Code, GitHub	Free

Total Estimated Cost

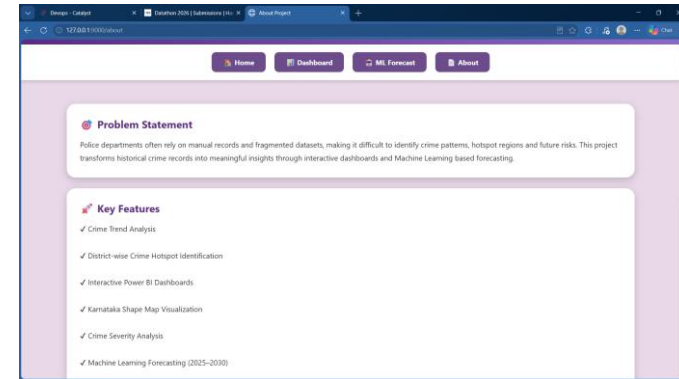
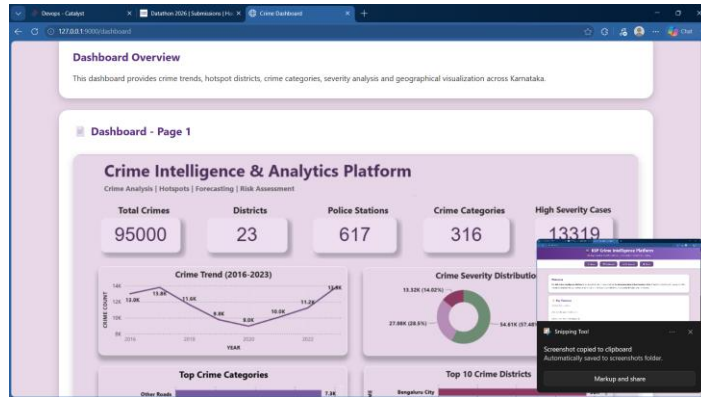
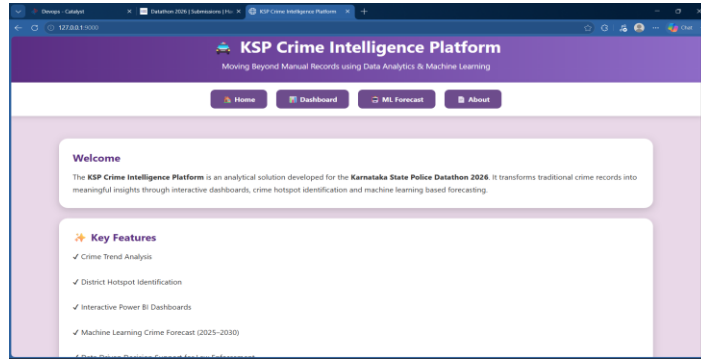
₹0 – ₹500

(Using free/open-source software and student/free-tier resources.)

- The proposed solution is highly cost-effective as it is built primarily using open-source technologies and free development platforms, making it affordable and easily scalable for government agencies.



Snapshots of the prototype :-



- The prototype provides an intuitive web interface for exploring crime statistics, interactive dashboards, and AI-driven crime prediction, enabling users to access insights efficiently.



Prototype Performance report/Benchmarking :-

Parameter	Result
Dataset Processed	102,735 Crime Records
Total Features	35 Columns
Database	MySQL
Dashboard	Power BI Interactive Dashboard
Prediction Model	Machine Learning (Scikit-learn)
Web Framework	Flask
Response Time	< 2 Seconds (Local Prototype)
User Interface	Responsive & Easy to Navigate
Status	Prototype Successfully Developed

❖ The prototype demonstrates the feasibility of an AI-driven crime analytics platform capable of transforming historical crime records into actionable intelligence for law enforcement agencies.



Project Resources:-

GitHub Public Repository :- <https://github.com/JashwanthRamadugu27/Crime-Intelligence-Analytics-Platform.git>

Demo Video Link:- https://drive.google.com/file/d/1M2Bgu5O-VrL1sI98nIt7A9qTxvTgzjVa/view?usp=drive_link

Deployed Link :- <https://crime-intelligence-analytics-platform-50044349180.development.catalystappsail.in>



Additional Details/Future Development:-

Future Enhancements

- **Real-Time Crime Monitoring** using live police data streams.
- **GIS-Based Interactive Crime Maps** for location-wise analysis.
- **Advanced AI Models** for higher prediction accuracy and risk assessment.
- **Mobile Application** for field officers to access insights on the go.
- **Secure Role-Based Authentication** for authorized police personnel.
- **Automated Crime Alerts & Notifications** for emerging crime hotspots.
- **Cloud Deployment & Scalability** to support statewide implementation.
- **Automated Report Generation** for faster administrative decision-making.

Potential Impact

- Enables **proactive policing** through predictive analytics.
- Improves **resource allocation** and operational efficiency.
- Supports **data-driven decision-making** for law enforcement.
- Enhances **public safety** with faster crime analysis and visualization.



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THANK YOU